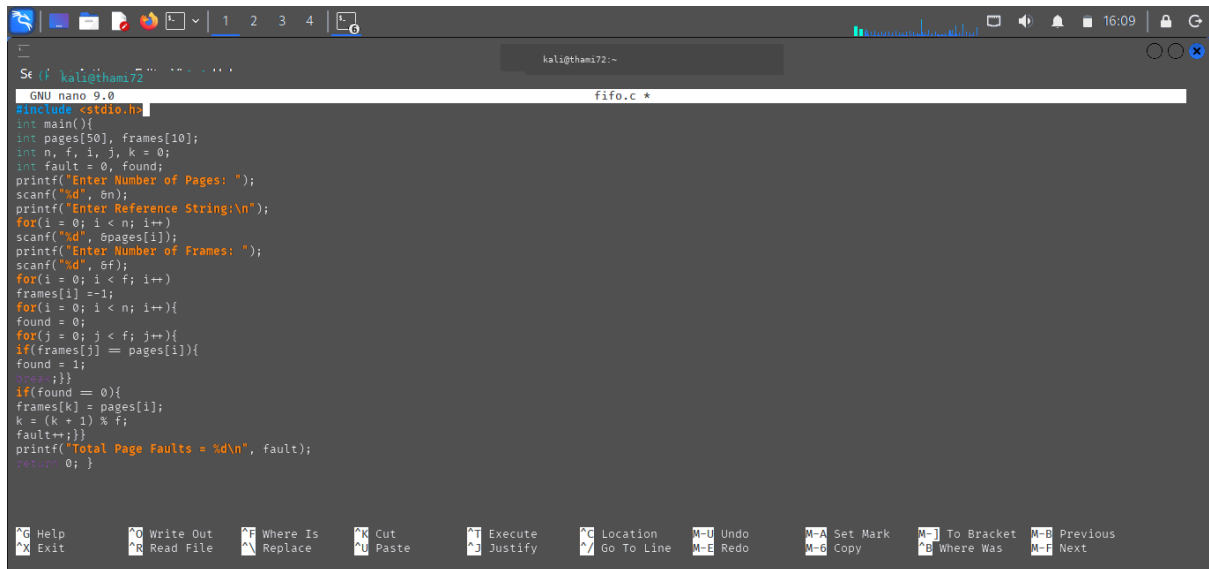
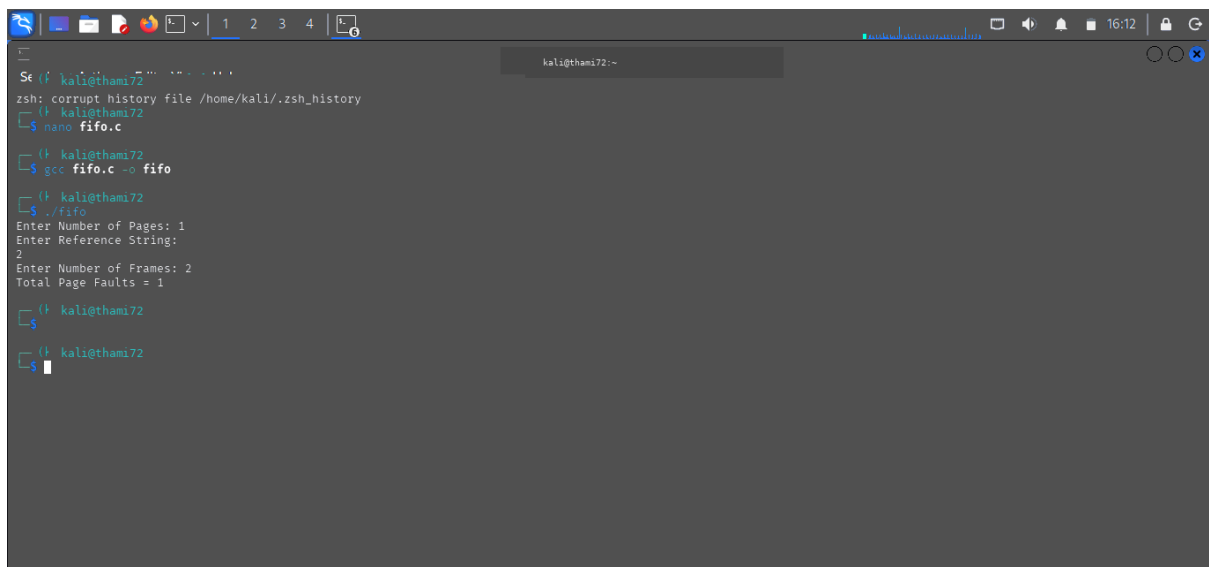


C PROGRAMMING : FIFO



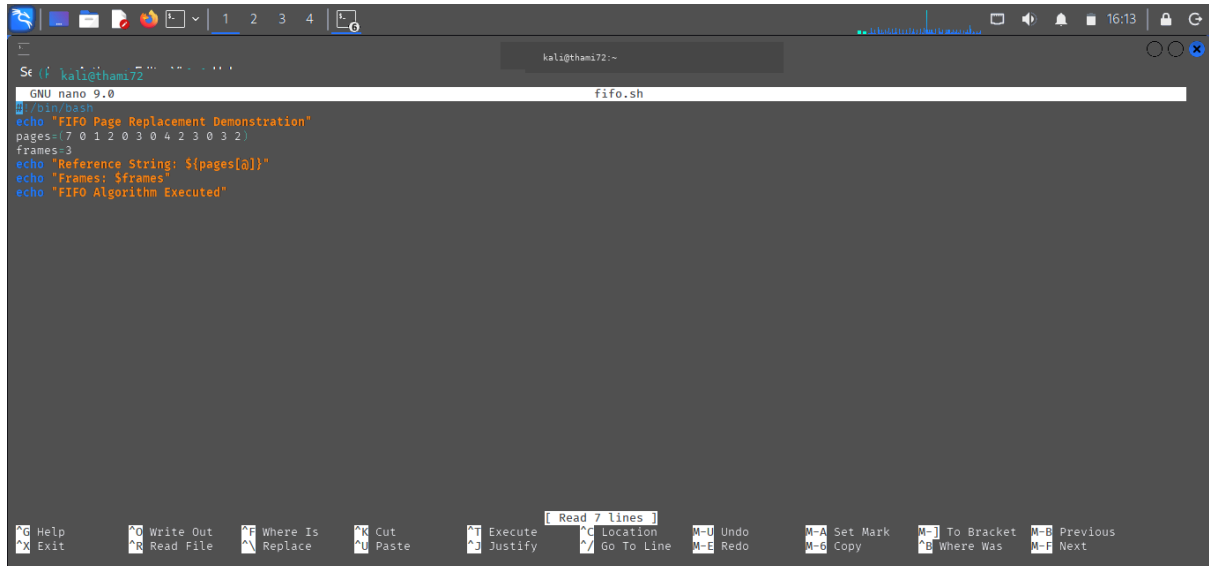
```
Se (f kali@thami72 ~)
GNU nano 9.0 fifo.c *
#include <stdio.h>
int main(){
    int pages[50], frames[10];
    int n, f, i, j, k = 0;
    int fault = 0, found;
    printf("Enter Number of Pages: ");
    scanf("%d", &n);
    printf("Enter Reference String:\n");
    for(i = 0; i < n; i++){
        scanf("%d", &pages[i]);
        printf("Enter Number of Frames: ");
        scanf("%d", &f);
        for(i = 0; i < f; i++){
            frames[i] = -1;
        }
        for(i = 0; i < n; i++){
            found = 0;
            for(j = 0; j < f; j++){
                if(frames[j] == pages[i]){
                    found = 1;
                }
            }
            if(found == 0){
                frames[k] = pages[i];
                k = (k + 1) % f;
                fault++;
            }
        }
        printf("Total Page Faults = %d\n", fault);
        return 0;
    }
```

OUTPUT :



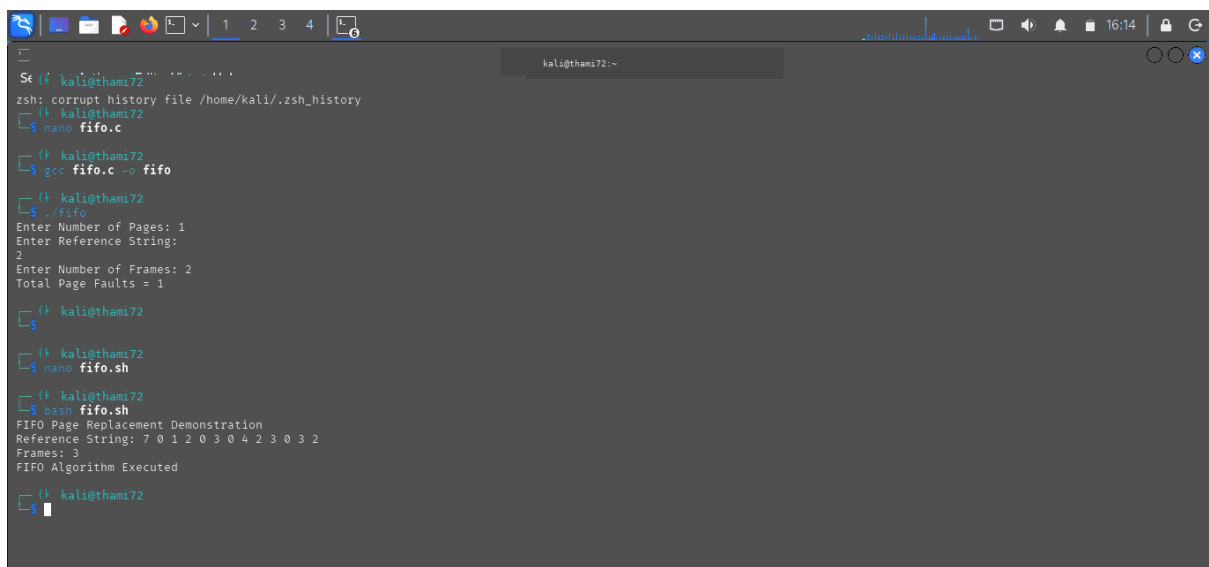
```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano fifo.c
kali@thami72 ~
$ gcc fifo.c -o fifo
kali@thami72 ~
$ ./fifo
Enter Number of Pages: 1
Enter Reference String:
2
Enter Number of Frames: 2
Total Page Faults = 1
kali@thami72 ~
$
```

SHELL PROGRAMMING : FIFO



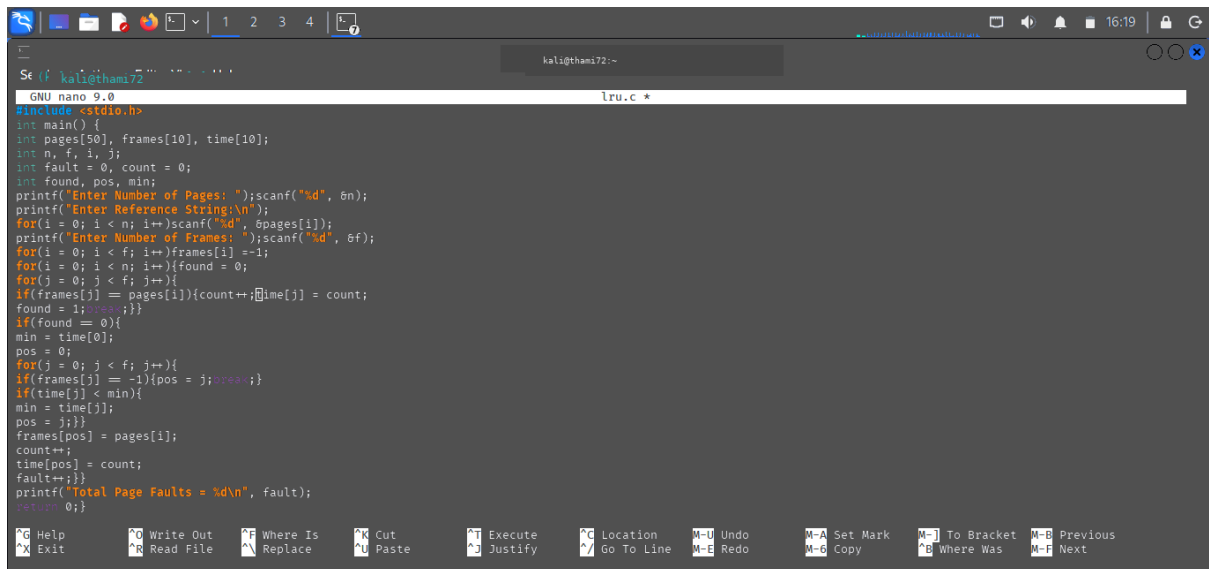
```
GNU nano 9.0 fifo.sh
#!/bin/bash
echo "FIFO Page Replacement Demonstration"
pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)
frames=3
echo "Reference String: ${pages[@]}"
echo "Frames: $frames"
echo "FIFO Algorithm Executed"
```

OUTPUT :



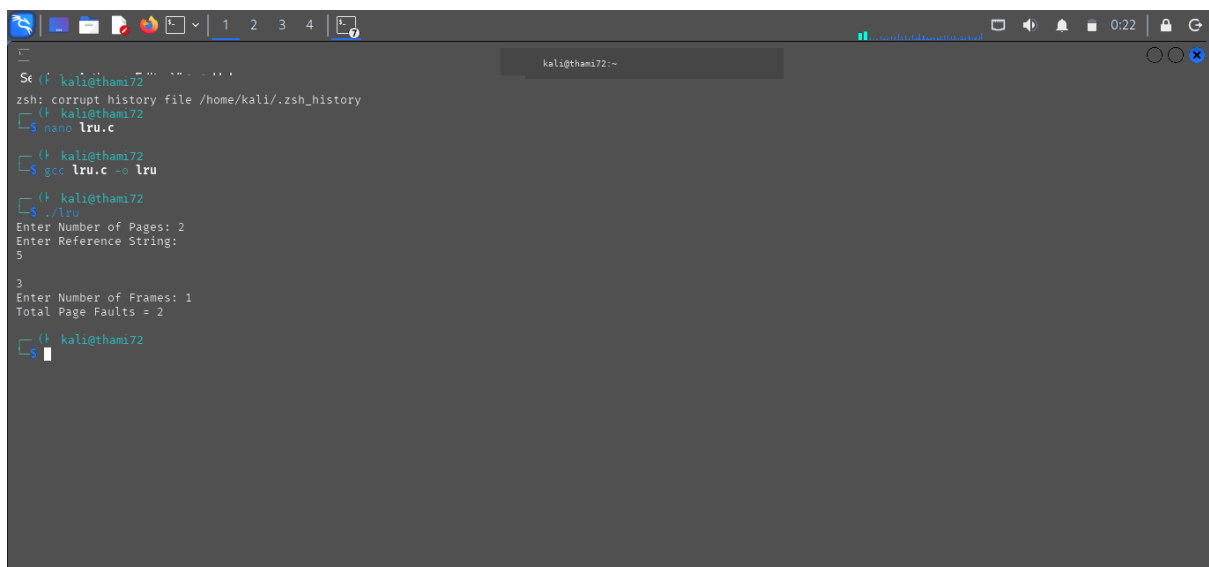
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano fifo.c
kali@thami72:~$ gcc fifo.c -o fifo
kali@thami72:~$ ./fifo
Enter Number of Pages: 1
Enter Reference String:
2
Enter Number of Frames: 2
Total Page Faults = 1
kali@thami72:~$ nano fifo.sh
kali@thami72:~$ bash fifo.sh
FIFO Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Frames: 3
FIFO Algorithm Executed
kali@thami72:~$
```

C PROGRAMMING :LRU



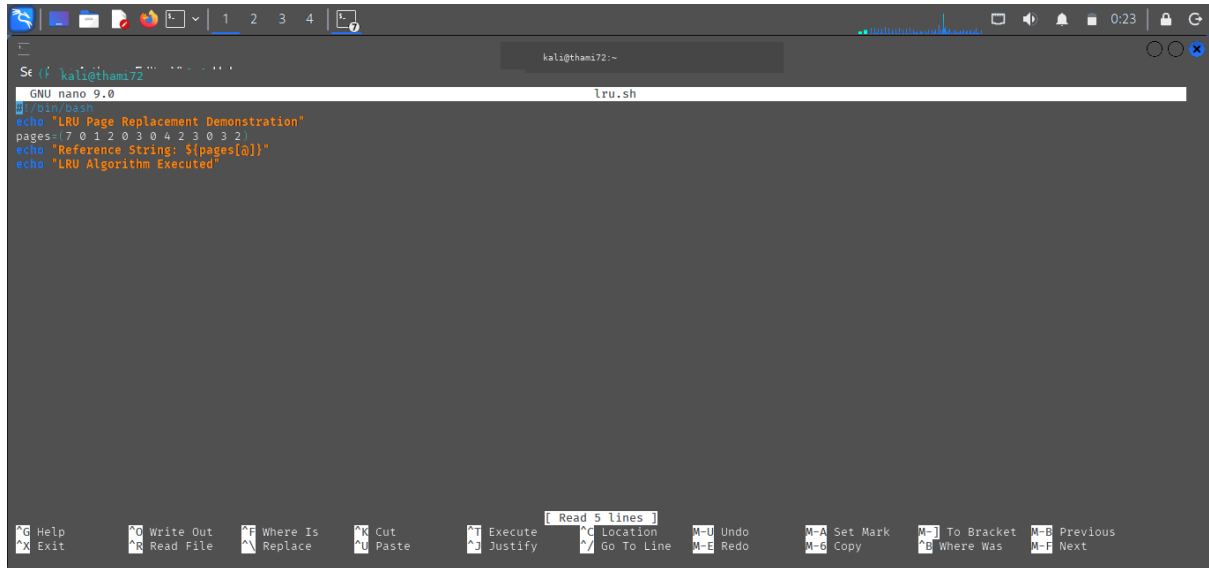
```
Se (f kali@thami72 ~)
GNU nano 9.0 lru.c *
#include <stdio.h>
int main() {
    int pages[50], frames[10], time[10];
    int n, f, i, j;
    int fault = 0, count = 0;
    int found, pos, min;
    printf("Enter Number of Pages: ");scanf("%d", &n);
    printf("Enter Reference String:\n");
    for(i = 0; i < n; i++)scanf("%d", &pages[i]);
    printf("Enter Number of Frames: ");scanf("%d", &f);
    for(i = 0; i < f; i++)frames[i] = -1;
    for(i = 0; i < n; i++){found = 0;
    for(j = 0; j < f; j++){
        if(frames[j] == pages[i]){count++;time[j] = count;
        found = 1;break;}}
    if(found == 0){
        min = time[0];
        pos = 0;
        for(j = 0; j < f; j++){
            if(frames[j] == -1){pos = j;break;}
            if(time[j] < min){
                min = time[j];
                pos = j;}}
        frames[pos] = pages[i];
        count++;
        time[pos] = count;
        fault++;}}
    printf("Total Page Faults = %d\n", fault);
    return 0;}
```

OUTPUT :



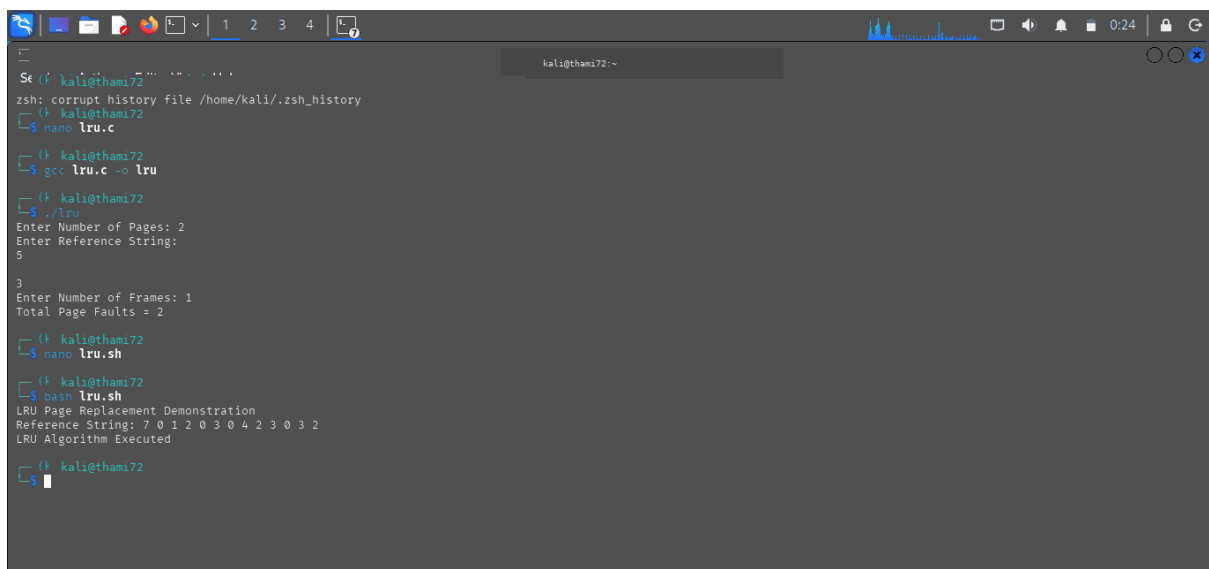
```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano lru.c
kali@thami72 ~
$ gcc lru.c -o lru
kali@thami72 ~
$ ./lru
Enter Number of Pages: 2
Enter Reference String:
5
3
Enter Number of Frames: 1
Total Page Faults = 2
kali@thami72 ~
$
```

SHELL PRROGRAMMING : LRU



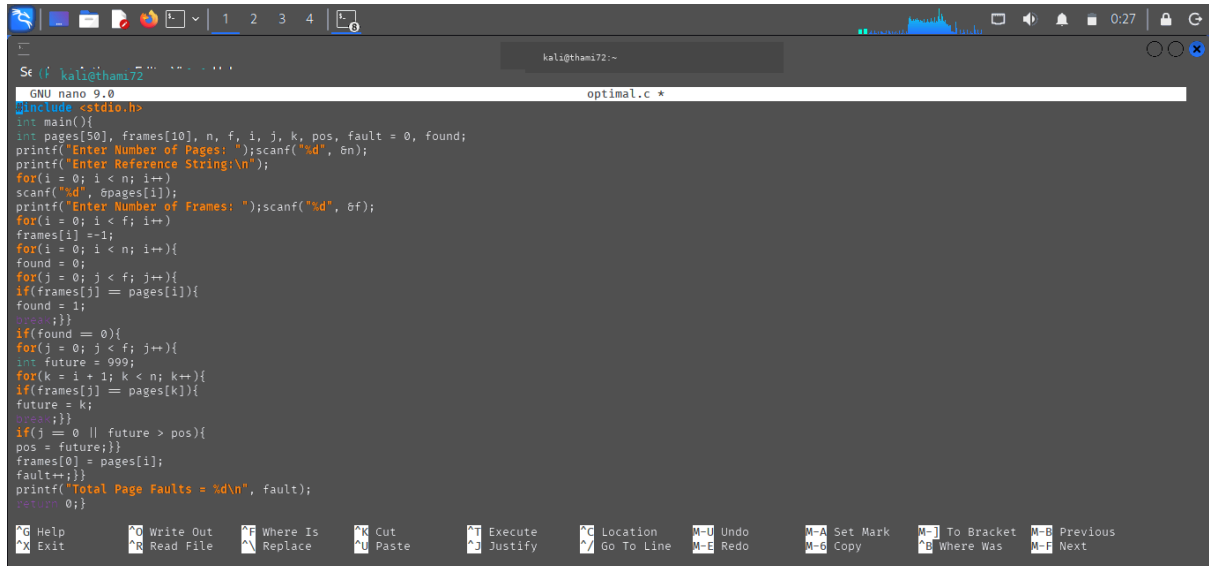
```
GNU nano 9.0 lru.sh
#!/bin/bash
echo "LRU Page Replacement Demonstration"
pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)
echo "Reference String: ${pages[@]}"
echo "LRU Algorithm Executed"
```

OUTPUT :



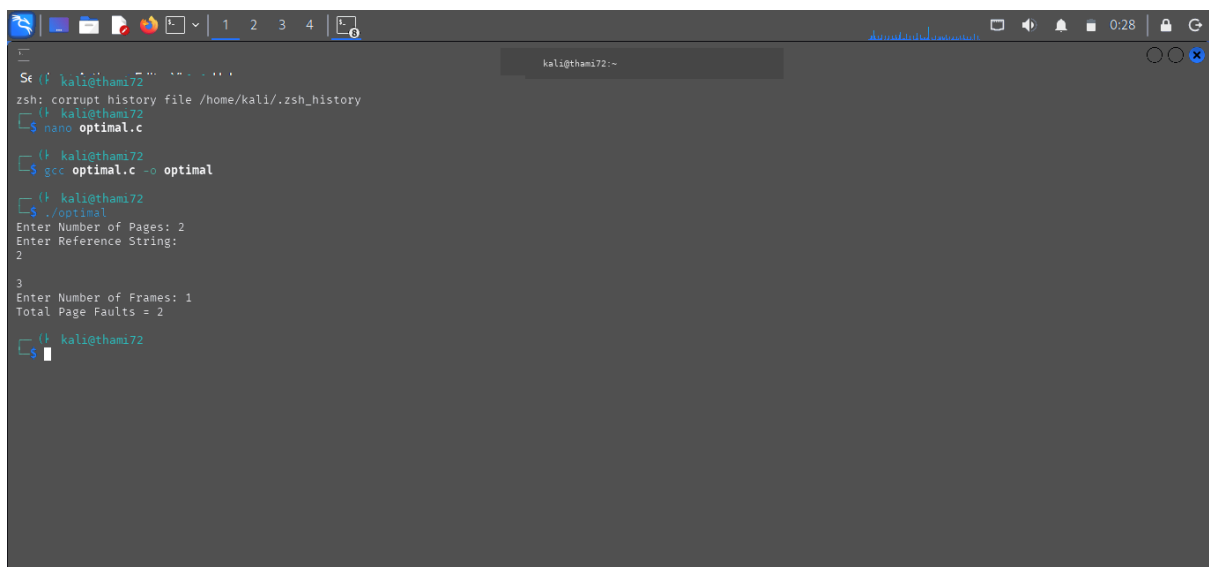
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
└─$ nano lru.c
└─$ gcc lru.c -o lru
└─$ ./lru
Enter Number of Pages: 2
Enter Reference String:
5
3
Enter Number of Frames: 1
Total Page Faults = 2
└─$ nano lru.sh
└─$ bash lru.sh
LRU Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
LRU Algorithm Executed
└─$
```

C PROGRAMMING :OPTIMAL PAGE



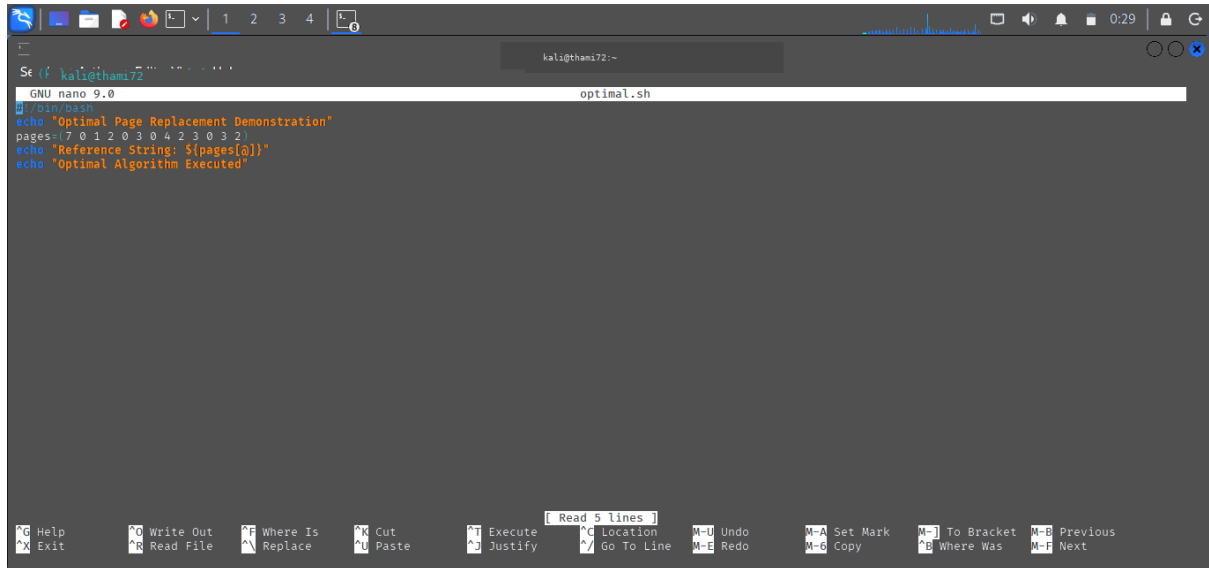
```
GNU nano 9.0 optimal.c *
#include <stdio.h>
int main(){
    int pages[50], frames[10], n, f, i, j, k, pos, fault = 0, found;
    printf("Enter Number of Pages: ");scanf("%d", &n);
    printf("Enter Reference String:\n");
    for(i = 0; i < n; i++){
        scanf("%d", &pages[i]);
    }
    printf("Enter Number of Frames: ");scanf("%d", &f);
    for(i = 0; i < f; i++){
        frames[i] = -1;
    }
    for(i = 0; i < n; i++){
        found = 0;
        for(j = 0; j < f; j++){
            if(frames[j] == pages[i]){
                found = 1;
                break;
            }
        }
        if(found == 0){
            for(j = 0; j < f; j++){
                int future = 999;
                for(k = i + 1; k < n; k++){
                    if(frames[j] == pages[k]){
                        future = k;
                        break;
                    }
                }
                if(j == 0 || future > pos){
                    pos = future;
                }
            }
            frames[0] = pages[i];
            fault++;
        }
    }
    printf("Total Page Faults = %d\n", fault);
    return 0;
}
```

OUTPUT :



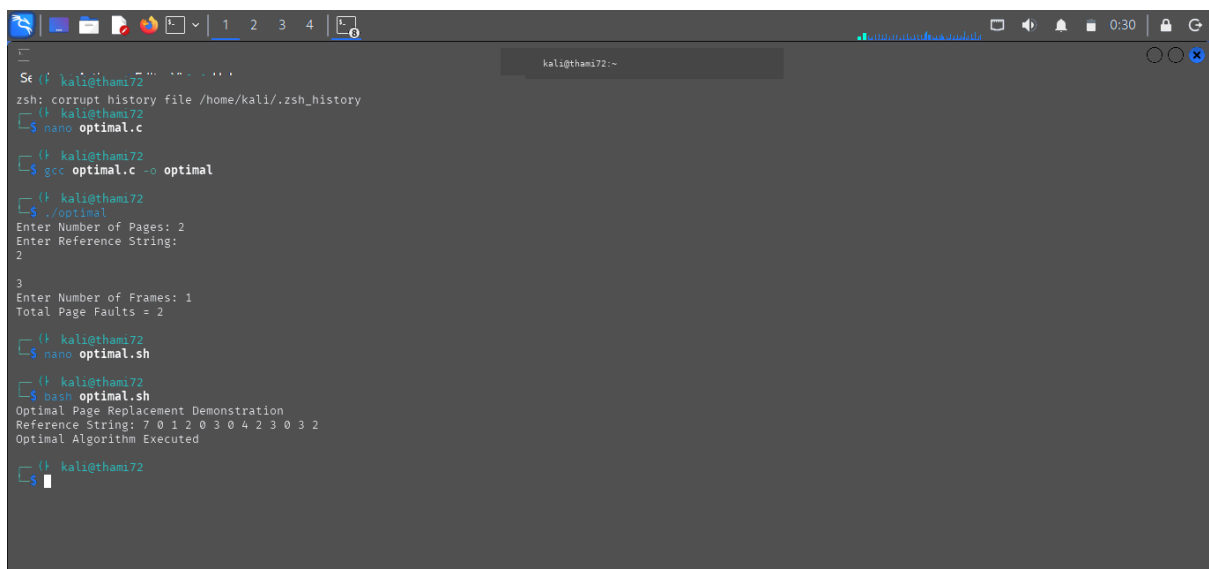
```
kali@thami72:~$ gcc optimal.c -o optimal
kali@thami72:~$ ./optimal
Enter Number of Pages: 2
Enter Reference String:
2
3
Enter Number of Frames: 2
Total Page Faults = 2
kali@thami72:~$
```

SHELL PROGRAMMING :



```
GNU nano 9.0 optimal.sh
#!/bin/bash
echo "Optimal Page Replacement Demonstration"
pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)
echo "Reference String: ${pages[@]}"
echo "Optimal Algorithm Executed"
```

OUTPUT :



```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano optimal.c
kali@thami72 ~
$ gcc optimal.c -o optimal
kali@thami72 ~
$ ./optimal
Enter Number of Pages: 2
Enter Reference String:
2
3
Enter Number of Frames: 1
Total Page Faults = 2
kali@thami72 ~
$ nano optimal.sh
kali@thami72 ~
$ dash optimal.sh
Optimal Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Optimal Algorithm Executed
kali@thami72 ~
$
```